A Minimal Unified Geometric Framework for Multi-Domain Dynamics

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Abstract

We present SEFI, a minimal unified geometric framework capable of describing physical, biological, and cognitive systems within a single mathematical structure. SEFI introduces a warp-modified metric, a canonical operator set, and a unified evolution equation that generalizes geometric flow. We demonstrate that these components yield coherent cross-domain behavior and provide a compact foundation for future unified theories.

1 Main

Unification in physics typically concerns interactions or fields. SEFI proposes a geometric unification: diverse systems arise from manifold geometry modified by a warp field and acted upon by a small operator set. The resulting dynamics capture structure, coherence, and evolution across domains.

1.1 Warp-metric

The canonical warp-metric is

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \alpha W g_{\mu\nu} + \beta \nabla_\mu \nabla_\nu W.$$

1.2 Operator set

SEFI defines four operators: warp, tension, coherence, and correction. These operators encode generalized tendencies of systems.

1.3 Unified evolution

The unified evolution equation is

$$\partial_t \tilde{g}_{\mu\nu} = -2\kappa_G \left(R_{\mu\nu} - \tfrac{1}{2} R \tilde{g}_{\mu\nu}\right) + S_{\mu\nu}[W, T, C].$$

1.4 Cross-domain behavior

We show that SEFI reproduces:

- geometric modulation in physical systems,
- restoration, duplication, and aging in biological systems,
- coherence and contextual modulation in cognitive systems.

2 Conclusion

SEFI provides a compact, unified geometric foundation with cross-domain applicability. Its minimal structure makes it suitable for high-impact theoretical development.

References